

Mathematics II (part 1)

CONTINUOUS PROBABILITY DISTRIBUTION (part 2)



Ryan Fraser Kirwan

What are we going to learn?

Real world examples
of Continuous spaces

- Standard normal distribution example

Binomial
approximation

- Intro, definition and examples

Combining random
variables

- Intro, definition and examples

CONTINUOUS SPACES

STANDARD NORMAL DISTRIBUTION EXAMPLE

Continuous spaces

Standard normal distribution (real-world) example

Example: A machine fills bottles with 300ml of soft drinks. However, the actual quantity filled varies according to the normal distribution with $\mu = 298\text{ml}$ and $\sigma = 3\text{ml}$. What is the probability that an individual bottle contains less than 295ml?

Continuous spaces

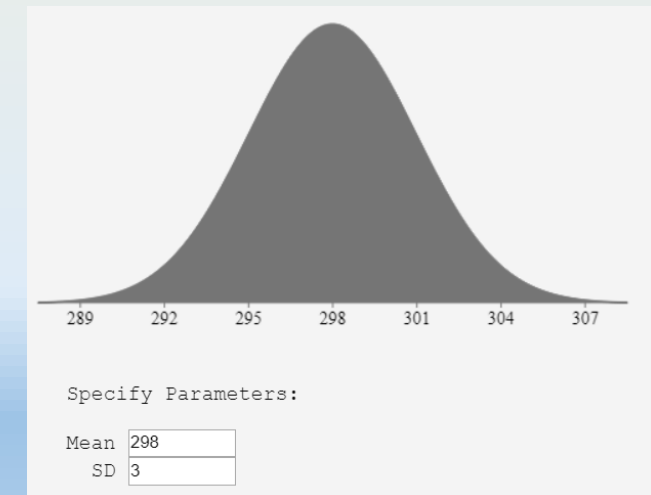
Standard normal distribution (real-world) example

Example: A machine fills bottles with 300ml of soft drinks. However, the actual quantity filled varies according to the normal distribution with $\mu = 298\text{ml}$ and $\sigma = 3\text{ml}$. What is the probability that an individual bottle contains less than 295ml?

Solution:

- Let X be the quantity of soft drinks in a bottle, $X \sim N(298, 3^2)$

What does
this mean?



http://onlinestatbook.com/2/calculators/normal_dist.html

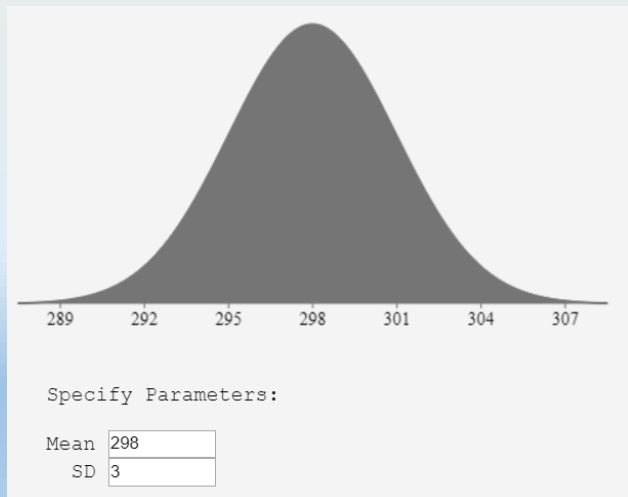
Continuous spaces

Standard normal distribution (real-world) example

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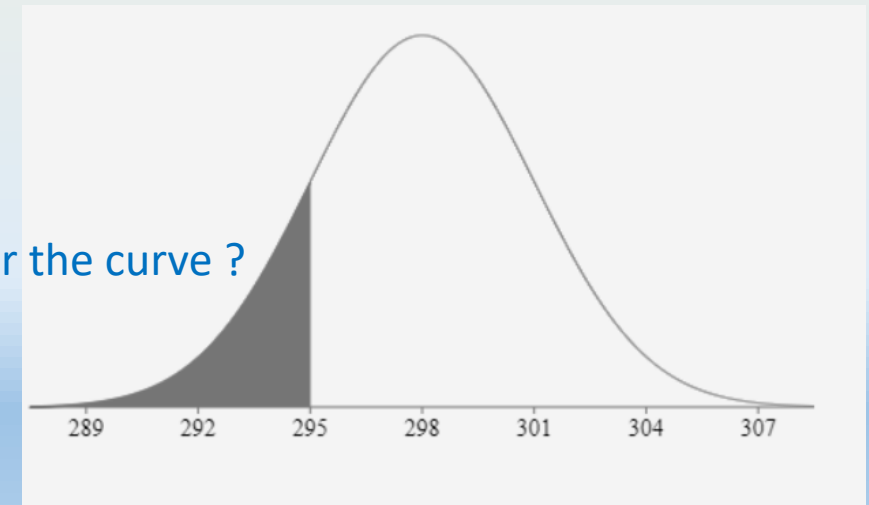
- Let X be the quantity of soft drinks in a bottle, $X \sim N(298, 3^2)$



$$P(X < 295)$$

?

The area under the curve ?



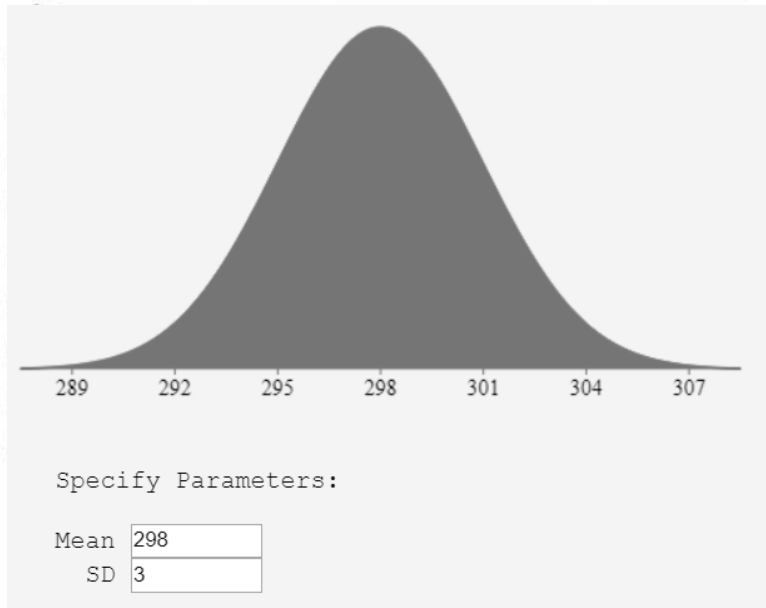
Continuous spaces

From normal to standard normal distributions

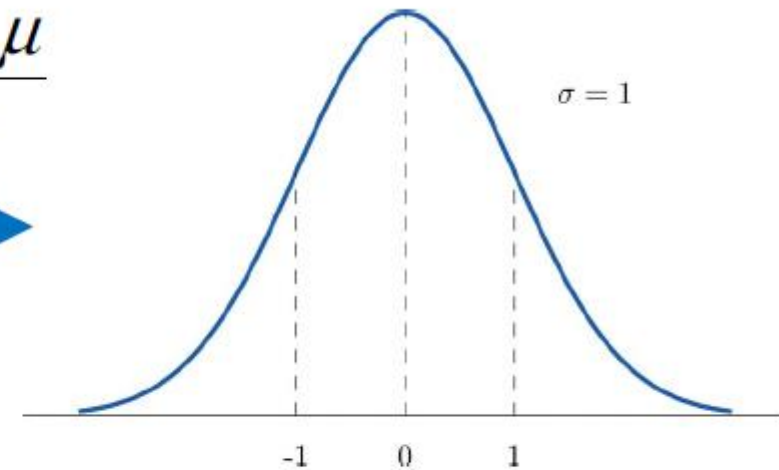
General normal distributions X transforming to standard normal distribution Z

$$X \sim N(298, 3^2)$$

$$Z \sim N(0, 1)$$



$$Z = \frac{X - \mu}{\sigma}$$



11

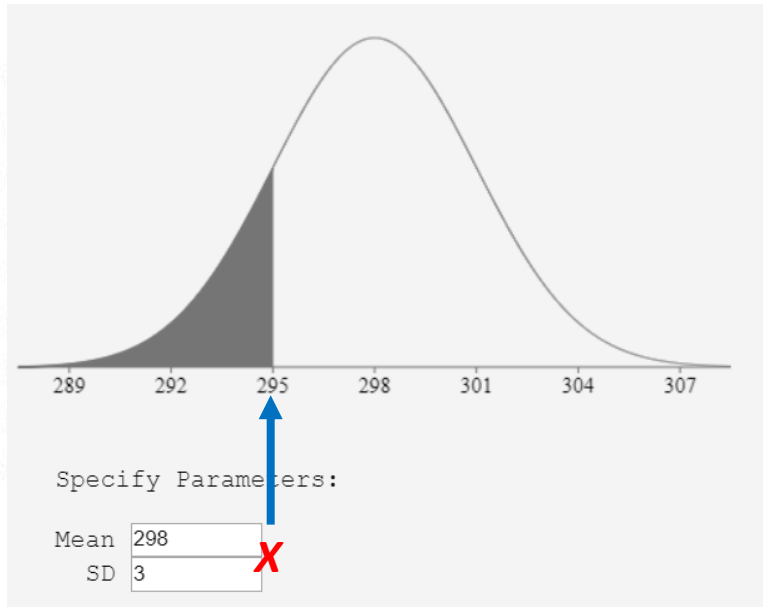
Continuous spaces

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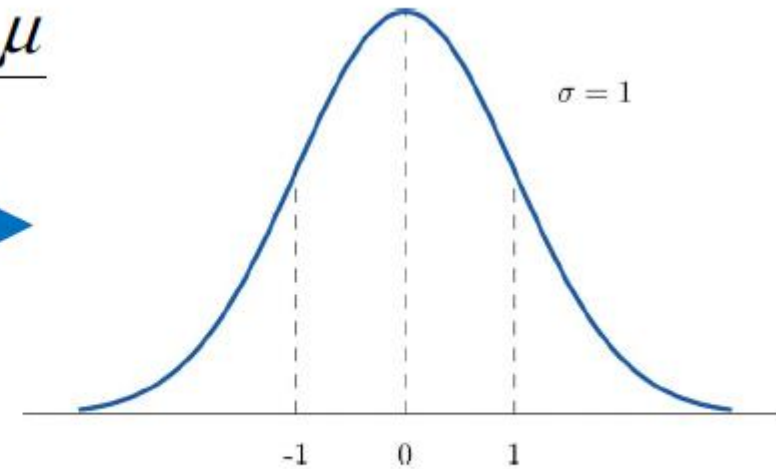
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11

$$P(X < 295) = P\left(Z < \frac{295 - 298}{3}\right) \leftarrow \text{to standard normal}$$

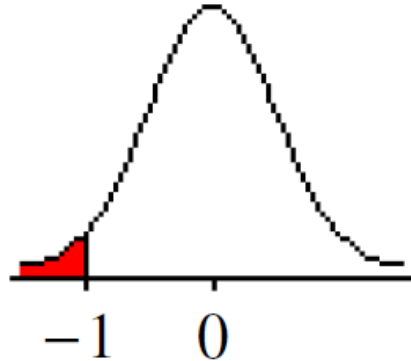
Continuous spaces

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Solution:

- Let X be the quantity of soft drinks in a bottle, $X \sim N(298, 3^2)$

$$\begin{aligned} P(X < 295) &= P\left(Z < \frac{295 - 298}{3}\right) \quad \leftarrow \text{to standard normal} \\ &= P(Z < -1) \end{aligned}$$


Continuous spaces

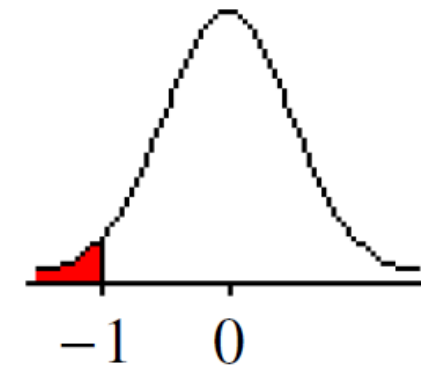
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- Let X be the quantity of soft drinks in a bottle, $X \sim N(298, 3^2)$

$$\begin{aligned} P(X < 295) &= P\left(Z < \frac{295 - 298}{3}\right) \quad \leftarrow \text{to standard normal} \\ &= P(Z < -1) \end{aligned}$$



BINOMIAL APPROXIMATION

Binomial approximation

Introduction

- Another useful application of a **normal distribution** is to use an **approximate binomial distribution** if the sample size n is large.
 - Let q be defined as: $q = (1 - p)$
- Given $X \sim B(n, p)$, when sample size n moves to values such that $np > 5$ and $nq > 5$, it can be observed that:

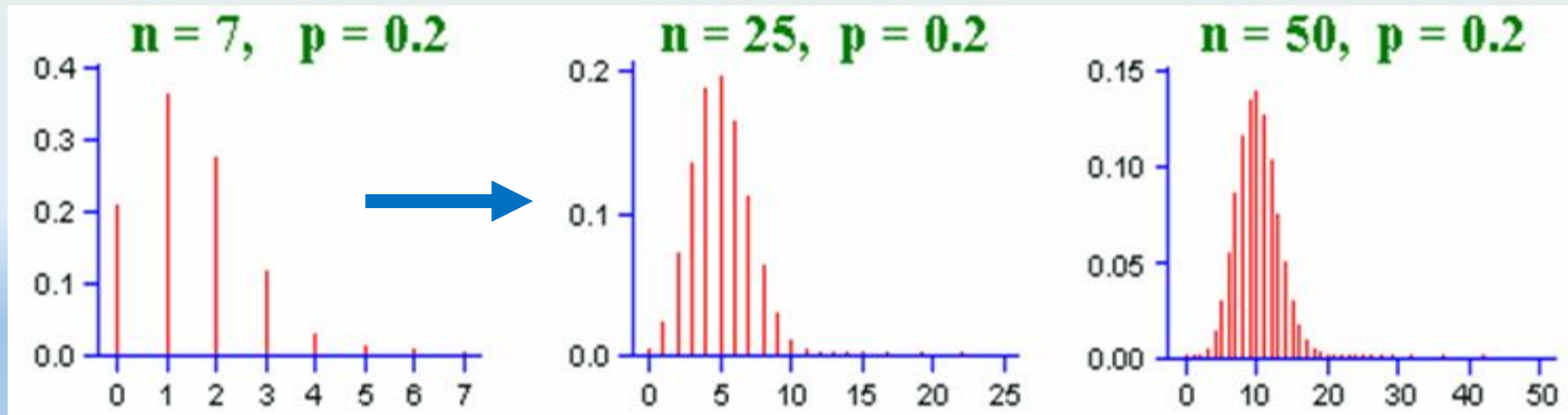
The mean and variance of $X \sim B(n, p)$ are:

$$E(X) = np$$

Mean, Expected value

$$\text{Var}(X) = np(1 - p)$$

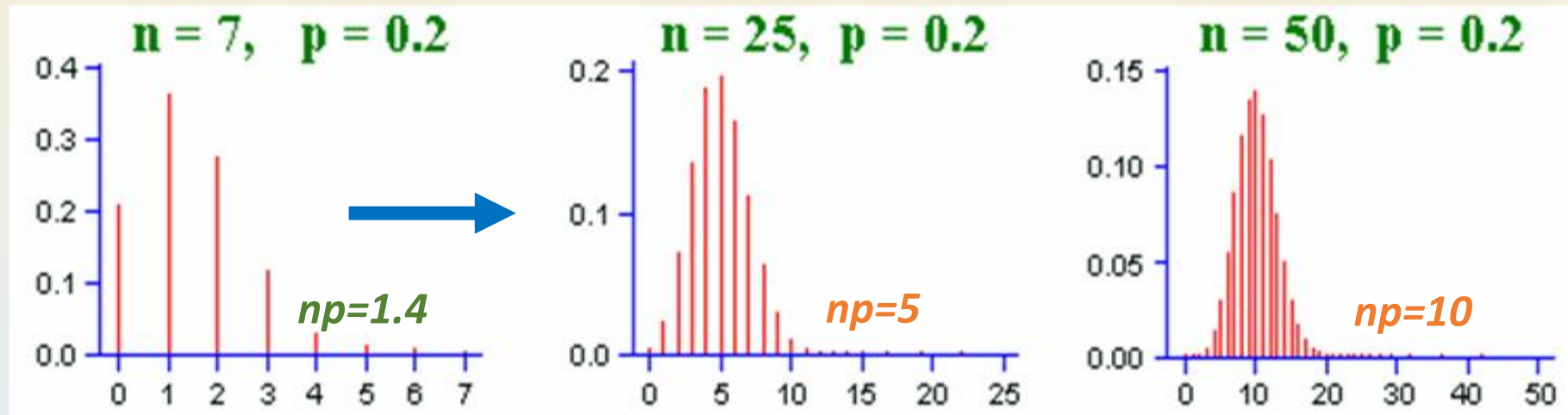
Variance



Binomial approximation

Introduction

Given $X \sim B(n, p)$, when sample size n is large such that $np > 5$ and $nq > 5$, it can be observed that:



$$B(n, p) \longrightarrow N(\mu, \sigma^2)$$

where $\mu = np$
 $\sigma^2 = npq$, $q = 1 - p$

Binomial approximation

Example

Example: In a particular school 60% of students are men and 40% are women. Given a sample of 50 students, what is the probability that more than half are women?



Binomial approximation

Example (without approximation)

Example: In a particular school 60% of students are men and 40% are women. Given a sample of 50 students, **what is the probability that more than half are women?**

Let X be the number of women, $X \sim B(50, 0.4)$

Without using **approximation**,

$$P(X > 25)$$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Binomial approximation

Example (without approximation)

Example: In a particular school 60% of students are men and 40% are women. Given a sample of 50 students, **what is the probability that more than half are women?**

Let X be the number of women, $X \sim B(50, 0.4)$

Without using **approximation**,

$$\begin{aligned} P(X > 25) &= P(X = 26) + P(X = 27) + \dots + P(X = 50) \\ &= \binom{50}{26} 0.4^{26} 0.6^{24} + \binom{50}{27} 0.4^{27} 0.6^{23} + \dots \\ &\quad + \binom{50}{50} 0.4^{50} 0.6^0 \\ &= 0.0573 \end{aligned}$$

Binomial approximation

Example (with approximation)

Solution:

Since

$$np = 50 \times 0.4 = 20 > 5$$

$$nq = 50 \times 0.6 = 30 > 5$$

Binomial approximation

Example (with approximation)

Solution:

Since $np = 50 \times 0.4 = 20 > 5$
 $nq = 50 \times 0.6 = 30 > 5$

We can approximate X with normal distribution $Y \sim N(\mu, \sigma^2)$

where $\mu = np = 20$
 $\sigma^2 = npq = 50 \times 0.4 \times 0.6 = 12$

Binomial approximation

Example (with approximation)

If $P(X = a)$ use $P(a-0.5 < X < a+0.5)$
If $P(X > a)$ use $P(X > a + 0.5)$
If $P(X \leq a)$ use $P(X < a + 0.5)$
If $P(X < a)$ use $P(X < a - 0.5)$
If $P(X \geq a)$ use $P(X > a - 0.5)$

Continuity correction is used to account for the rounding of continuous values to discrete values. E.g., the continuous value 45.5 would be considered 46 as a discrete value.

Solution:

Since

$$np = 50 \times 0.4 = 20 > 5$$

$$nq = 50 \times 0.6 = 30 > 5$$

We can approximate X with normal distribution $Y \sim N(\mu, \sigma^2)$

where

$$\mu = np = 20$$

$$\sigma^2 = npq = 50 \times 0.4 \times 0.6 = 12$$

$$P(X > \boxed{25}) = P(Y > 25.5)$$

← continuity correction

$\boxed{a}$

Binomial approximation

Example (with approximation)

Solution:

Since

$$np = 50 \times 0.4 = 20 > 5$$

$$nq = 50 \times 0.6 = 30 > 5$$

We can approximate X with normal distribution $Y \sim N(\mu, \sigma^2)$

where

$$\mu = np = 20$$

$$\sigma^2 = npq = 50 \times 0.4 \times 0.6 = 12$$

$$P(X > 25) = P(Y > 25.5) \quad \leftarrow \text{continuity correction}$$

$$= P\left(Z > \frac{25.5 - 20}{\sqrt{12}}\right) \quad \leftarrow \text{to standard normal}$$

$$= P(Z > 1.59)$$

$$Z = \frac{X - \mu}{\sigma}$$

Binomial approximation

Example (with approximation)

Solution:

Since

$$np = 50 \times 0.4 = 20 > 5$$

$$nq = 50 \times 0.6 = 30 > 5$$

We can approximate X with normal distribution $Y \sim N(\mu, \sigma^2)$

where

$$\mu = np = 20$$

$$\sigma^2 = npq = 50 \times 0.4 \times 0.6 = 12$$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9978	.9979	.9979	.9980	.9981	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986

$$P(X > 25) = P(Y > 25.5) \quad \leftarrow \text{continuity correction}$$

$$= P\left(Z > \frac{25.5 - 20}{\sqrt{12}}\right) \quad \leftarrow \text{to standard normal}$$

$$= P(Z > 1.59)$$

$$= 1 - P(Z \leq 1.59)$$

$$= 1 - 0.9441 = 0.0559 \quad \leftarrow \text{from Z-table}$$

Actual calc = 0.0573

COMBINING RANDOM VARIABLES

Combining random variables

Introduction

Adding or subtracting random variables to each other

Add the expected values

- $E(X + Y) = E(X) + E(Y)$

Add the variances

- $Var(X + Y) = Var(X) + Var(Y)$

Subtracted the expected values

- $E(X - Y) = E(X) - E(Y)$

Add the variances!!!!

- $Var(X - Y) = Var(X) + Var(Y)$

Combining random variables

Definitions

Theorems: Given discrete or continuous random variables X, Y and constants a, b :

Theorems: Given discrete or continuous random variables X, Y and constants a, b :

$$E(aX \pm b) = aE(X) \pm b$$

$$\text{Var}(aX \pm b) = a^2 \text{Var}(X)$$

$$E(aX \pm bY) = aE(X) \pm bE(Y)$$

$$\text{Var}(aX \pm bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$$

if X and Y are independent

Note:

- If X and Y are normal, then the resultant $aX \pm bY$ is also normal.
- If X and Y are binomial, then the resultant $aX + bY$ is binomial, only if they have the same success probability p .

Combining random variables

Example

A company offers a buyer's insurance policy that costs a customer \$70 dollar. The company will make a payout of \$10,000 dollars to the customer if they are victim to fraud in that year. The company set this price based on the probability of fraud being 0.002.

Let X represent the **company's profit** from one of these policies. The table below displays the probability distribution and summary statistics for X .

	No Fraud	Fraud
$X = \text{Profit } (\$)$	70	-9,930
$P(X)$	0.998	0.002
	Mean	\$50
	Standard Deviation	~ 447

Combining random variables

Example (shifting the scale)

What if we add \$10 more expected profit value?

Shift profit = $X+10$ hence,

$$\begin{aligned} I.E.: \mu_{new} &= \mu_{prev} + 10 \\ &= 50 + 10 \\ &= 60 \end{aligned}$$

$$E(a*X \pm b) = a*E(X) \pm b; a = 1, b = 10;$$

$$\begin{aligned} E(1*X + 10) &= E(Y) = 1*E(X) + 10 \\ &= 50 + 10 \\ &= 60 \end{aligned}$$

Theorems: Given discrete or continuous random variables X , Y and constants a , b :

$$\begin{aligned} E(aX \pm b) &= aE(X) \pm b \\ \text{Var}(aX \pm b) &= a^2\text{Var}(X) \end{aligned}$$

Adding 10 does not impact **standard deviation**. Adding or subtracting a constant only shifts data, which doesn't affect the spread.

$$\sigma_Y = \sigma_X \approx 447$$

Combining random variables

Example (combining two variables)

Example: Suppose the time for Meg to travel from home to airport is normally distributed with $\mu = 25$ min and $\sigma = 3$ min, and the time for her to clear the airport screening to boarding the plane is normally distributed with $\mu = 15$ min and $\sigma = 2$ min.

Combining random variables

Example (combining two variables)

Example: Suppose the time for Meg to travel from home to airport is normally distributed with $\mu = 25$ min and $\sigma = 3$ min, and the time for her to clear the airport screening to boarding the plane is normally distributed with $\mu = 15$ min and $\sigma = 2$ min.

- Given that Meg leaves home 45 mins before the last call to board. What is the probability that she will miss her flight?
- In other words, we want to know probability of total time ((Time to airport + Time to screening) > 45)

$$\begin{aligned}E(aX \pm bY) &= aE(X) \pm bE(Y) \\ \text{Var}(aX \pm bY) &= a^2\text{Var}(X) + b^2\text{Var}(Y) \\ &\text{if } X \text{ and } Y \text{ are independent}\end{aligned}$$



Combining random variables

Example (combining two variables)

Example: Suppose the time for Meg to travel from home to airport is normally distributed with $\mu = 25$ min and $\sigma = 3$ min, and the time for her to clear the airport screening to boarding the plane is normally distributed with $\mu = 15$ min and $\sigma = 2$ min.

Solution:

Let X be the time to travel from home to airport, $X \sim N(25, 3^2)$, and
 Y be the time to clear airport screening to boarding, $Y \sim N(15, 2^2)$.

Combining random variables

Example (combining two variables)

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Solution:

Let X be the time to travel from home to airport, $X \sim N(25, 3^2)$, and
 Y be the time to clear airport screening to boarding, $Y \sim N(15, 2^2)$.

Let $T = X + Y$, and assume X and Y are independent, then

$$E(T) = E(X + Y) = E(X) + E(Y) = 25 + 15 = 40$$

$$Var(T) = Var(X + Y) = Var(X) + Var(Y) = 3^2 + 2^2 = 13$$

Combining random variables

Example (combining two variables = New distribution)

Since combination of normal variables is still **normal** distribution, we have

$$T \sim N(40, 13)$$

Probability that she will miss her flight,

$$\begin{aligned} P(T > 45) &= P\left(Z > \frac{45 - 40}{\sqrt{13}}\right) \quad \leftarrow \text{to standard normal} \\ &= P(Z > 1.39) \\ &= 1 - P(Z < 1.39) \\ &= 1 - 0.9177 = 0.0823 \quad \leftarrow \text{from Z-table} \end{aligned}$$

SUMMARY

- ✓ Real world examples of Continuous spaces
- ✓ **Binomial approximation**
 - ✓ Approximation of **Binomial Distributions** to Normal Distribution
- ✓ **Combining Random Variables**

SUMMARY

Some useful formulae:

Continuous random variable:

- mean $\mu = \int_{-\infty}^{\infty} xf(x)dx$
- variance $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$
- standard deviation $\sigma = \sqrt{\sigma^2}$

Normal random variable $X \sim N(\mu, \sigma^2)$:

- mean μ
- variance σ^2

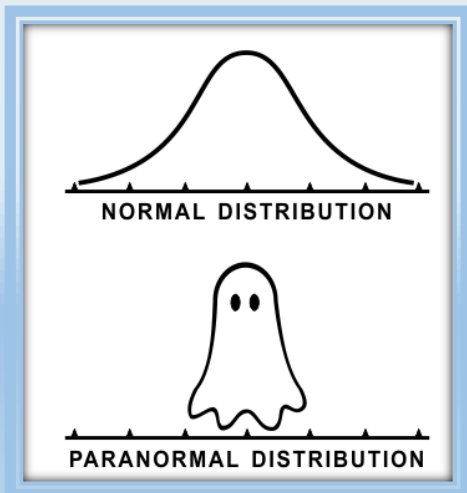
Using normal random variable Y to approximate binomial random variable X

- $X \sim B(n, p) \rightarrow Y \sim N(np, np(1-p))$

Theorems on combination of random variables X, Y

Mathematics II (part 1)

SAMPLES & SAMPLING DISTRIBUTION (part 1)



- Ryan Kirwan part 1
- Teh Yong Liang part 2

Samples & Sampling distributions

Random recap

Random Variables: Discrete and Continuous

Discrete Random Variable:

- mean $\mu = \sum x_i P(X = x_i)$
- variance $\sigma^2 = \sum (x_i - \mu)^2 P(X = x_i)$
- standard deviation $\sigma = \sqrt{\sigma^2}$

Binomial Random Variable

$$X \sim B(n, p)$$

- Probability Distribution
- Mean
- Variance

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

- mean $\mu = np$
- variance $\sigma^2 = np(1-p)$

Continuous random variable:

- mean $\mu = \int_{-\infty}^{\infty} xf(x)dx$
- variance $\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x)dx$
- standard deviation $\sigma = \sqrt{\sigma^2}$

Normal random variable $X \sim N(\mu, \sigma^2)$:

- mean μ
- variance σ^2

Using normal random variable Y to approximate binomial random variable X

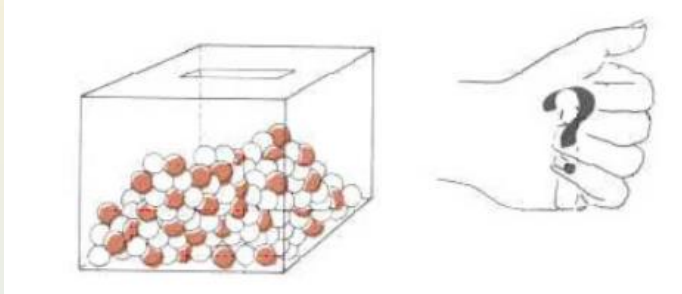
- $X \sim B(n, p) \rightarrow Y \sim N(np, np(1-p))$

Theorems on combination of random variables X, Y

Samples & Sampling distributions

Determining uncertainty with probability

Probability Statistics: Apply suitable probability distribution model for the random phenomenon (the box), predict the outcome (on hand)



e.g. given a fair die (the box), how likely to get a certain outcome (on hand)?

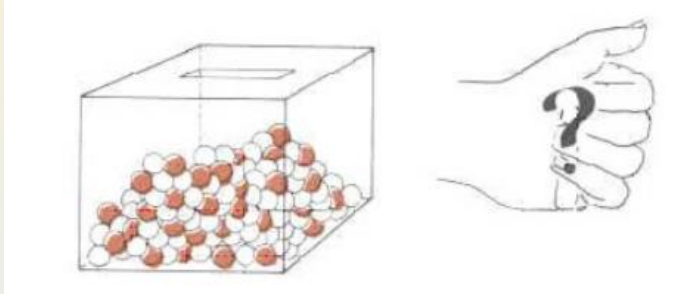
You are given the μ and σ of the population and then asked about the probability of a certain outcome.

What if μ and σ are unknown?

Samples & Sampling distributions

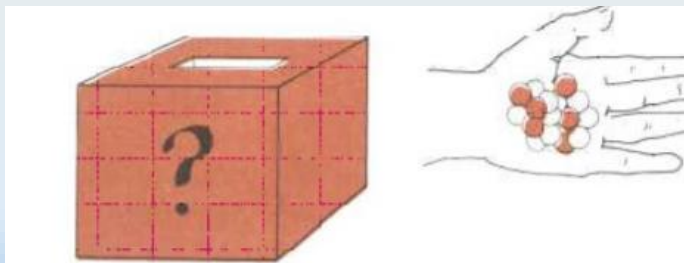
Determining uncertainty with probability

Probability Statistics: Apply suitable probability distribution model for the random phenomenon (the box), predict the outcome (on hand)



e.g. given a fair die (the box), how likely to get a certain outcome (on hand)?

Inferential Statistics: Given the outcome (on hand), infer the random phenomenon (the box) with the help of probability distribution model.



e.g. given a certain outcome has occurred (on hand), is the dice fair (the box)?

What are we going to learn?

Statistical studies

- Formulate questions
- Collect Data
- Answer question (by inference from data)

Types of studies

- Categorization
- Examples

Correlation vs Causation

- Example
- Difference

STATISTICAL STUDIES

Statistical Studies

How do we do inference?

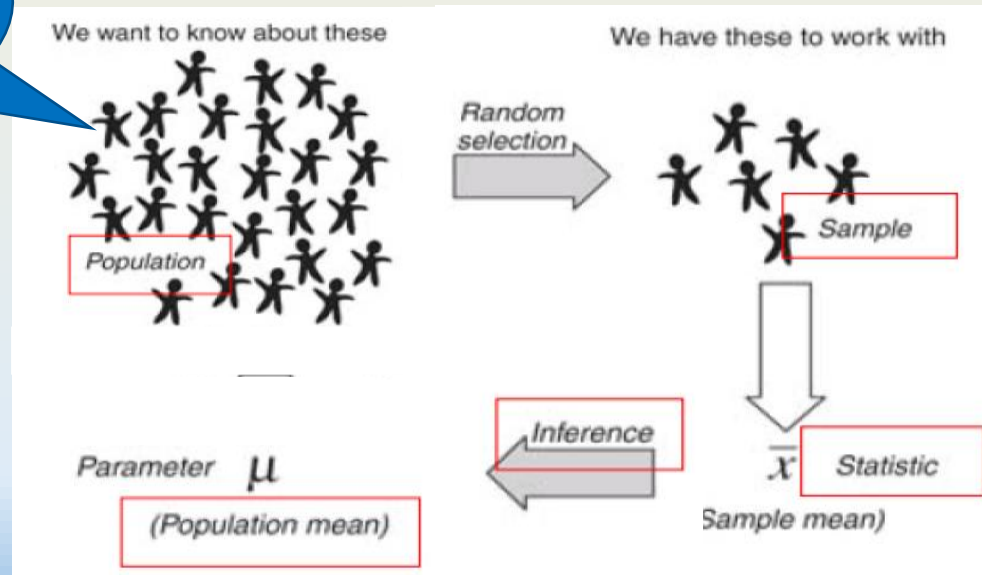
The purpose of **Inferential Statistics** is to obtain **sample statistics** to make **inferences** about **population parameters** in order to answer our questions.

The population parameter, e.g. mean μ , is unknown.

Conducting a census can be

Time-consuming

Impossible

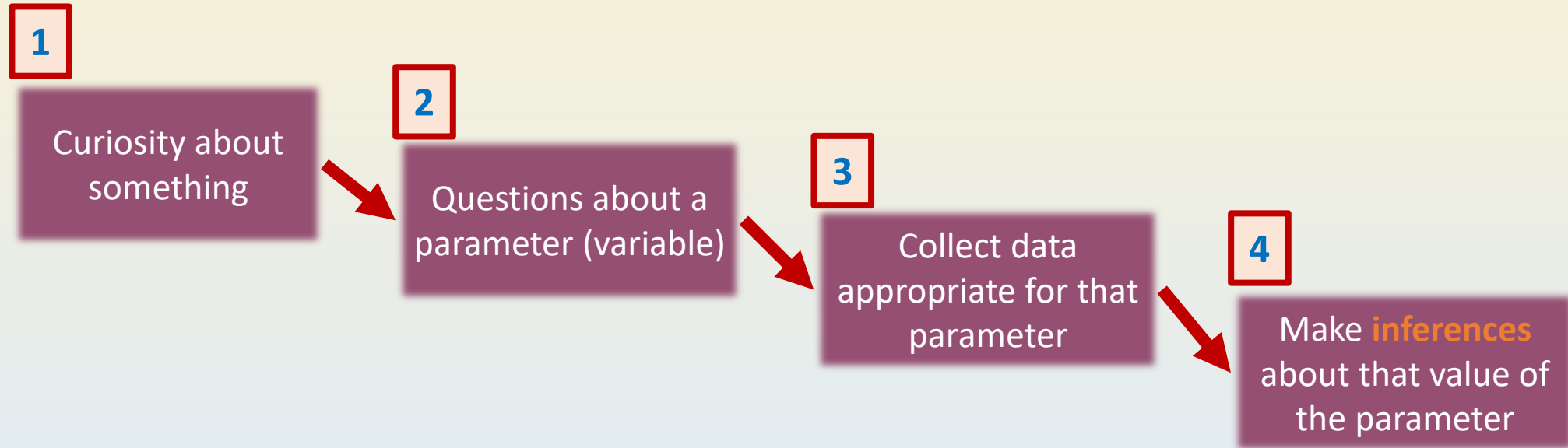


Inferential Statistics is about randomly selecting a representative sample to obtain **sample statistic**, e.g. calculate mean $\bar{x}$, to infer the unknown μ

Statistical Studies

Process of statistical studies

In general, the process for performing a statistical study comprises four steps:



Statistical Studies

First, what are statistical questions?

To answer statistical questions, you need to **collect data with variability**

Variability: the degree to which data points vary

For example, let us identify which of the following is a statistical question:

1. Am I sleepy?
Not statistical
2. How much time do my friends spend sleeping, on average, in a year?
Statistical
3. How many students are attending today's lecture?
Not statistical

What are statistical questions? (examples)

- Q • How heavy are elephants?
- How old are you?
- How old, on average, are the students who took the class this year?
- Do wolves weigh more than dogs?
- Is your cat heavier than that dog?
- How many teeth are in your dog's mouth?
- Do university students gain weight during their first year?

Statistical Studies

What are statistical questions? Answers (examples)

A

- **How heavy are elephants?**
- How old are you?
- **How old, on average, are the students who took the class this year?**
- **Do wolves weigh more than dogs?**
- Is your cat heavier than that dog?
- How many teeth are in your dog's mouth?
- **Do university students gain weight during their first year?**

Statistical Studies

Study design: What are valid approaches? (examples)

Curiosity: Do university students gain weight during their first year?

Consider how you would design a study to answer this.

Study design: What are valid approaches? (examples)

Curiosity: Do university students gain weight during their first year?

- I will take the difference between weights of a sample of students at end of the year and their weights at the beginning of the year
- I will measure the mean weight gain of all university students during the first year, and mean weight gain of all people of the same age during that same time but who do not go to university.
- I will measure the mean of weight gain of female students and the mean weight gain of male university students
- I will measure the mean weight of students and mean weight of faculty members to compare
- I will take proportion of the population of first-year university students who weigh more at the end of the year than they did at the beginning of the year.

Statistical Studies

Study design: What are valid approaches? Answers (examples)

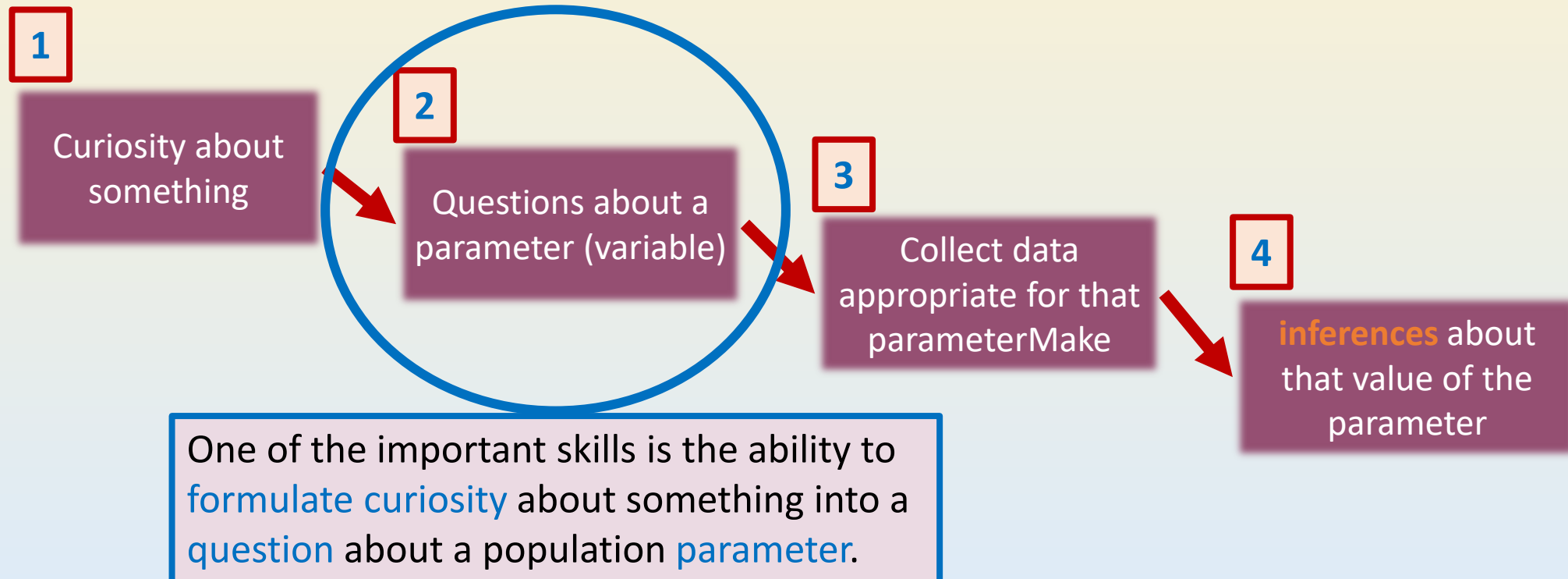
Curiosity: Do university students gain weight during their first year?

- I will take the difference between weights of a sample of students at end of the year and their weights at the beginning of the year
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- I will take proportion of the population of first-year university students who weigh more at the end of the year than they did at the beginning of the year.

Statistical Studies

Process of statistical studies

In general, the process for performing a statistical study comprises four steps:



Statistical Studies

Formulate a question into a parameter: quantitative variables

(for this module) We should be able to formulate our question into one of these three questions about population **parameters** as follows:

	Population parameter to be inferred	Sample statistic to collect	Notes
For Quantitative Variables			
One population mean	?	$\bar{x}$	Take <u>one measurement</u> from same sample
Population Mean of paired differences “ dependent ”	?	$\bar{d}$	Take <u>two</u> <u>measurements</u> from same sample
Population Mean of paired differences “ independent ”	?	$\bar{x}_1 - \bar{x}_2$	Take <u>two</u> <u>measurements</u> from two different independent samples

Statistical Studies

Formulate a question into a parameter: quantitative variables

(for this module) We should be able to formulate our question into one of these three questions about population **parameters** as follows:

	Population parameter to be inferred	Sample statistic to collect	Notes
For Quantitative Variables			
One population mean	μ	$\bar{x}$	Take <u>one measurement</u> from same sample
Population Mean of paired differences “ dependent ”	μ_d	$\bar{d}$	Take <u>two</u> <u>measurements</u> from same sample
Population Mean of paired differences “ independent ”	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	Take <u>two</u> <u>measurements</u> from two different independent samples

Formulate a question into a parameter: what will you infer?

Curiosity: Do university students gain weight during their first year?

parameter μ_d : **mean weight gain** (difference between weight at end of the year and weight at beginning of the year) during the first year for the population of university students.

parameter $\mu_1 - \mu_2$: where μ_1 = **mean weight gain** of all university students during the first year, and μ_2 = **mean weight gain** of all people of the same age during that same time but who do not go to university.

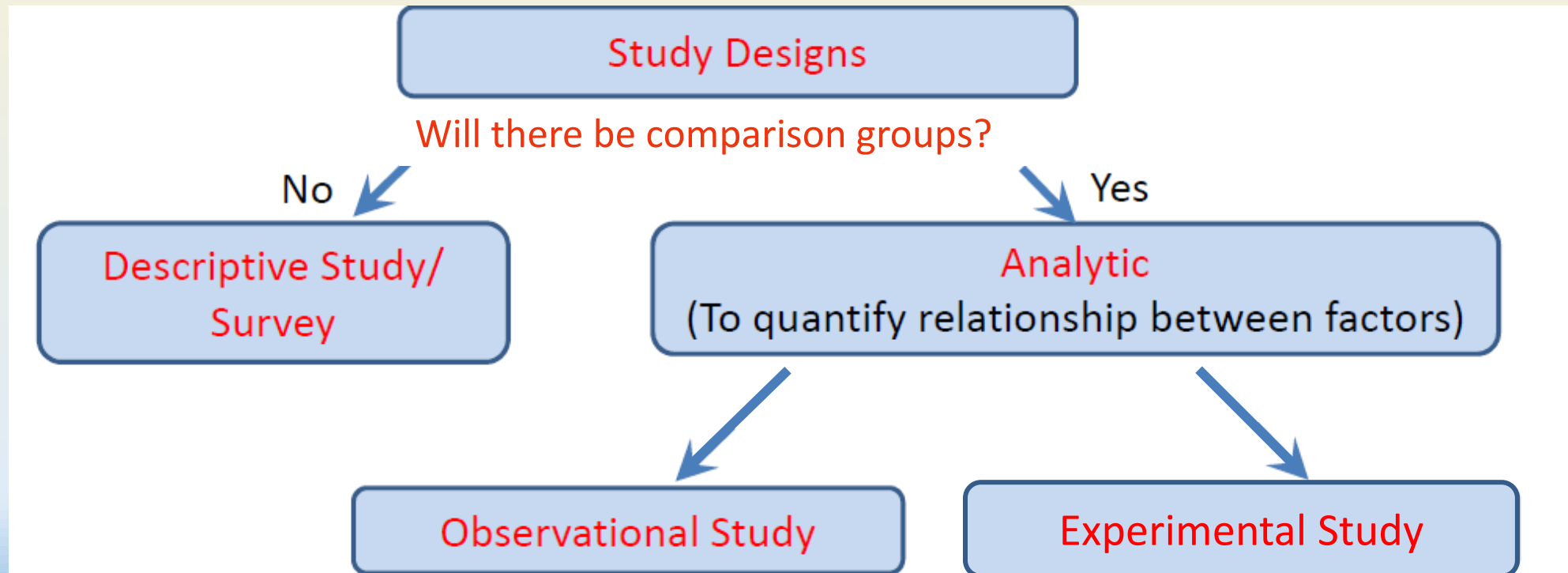
TYPES OF STATISTICAL STUDIES

Examples: Can meditation improve test scores?
Can drinking coffee help you live longer?

Types of statistical studies

What type of study to use?

Next, decide on the type of statistical studies to be conducted, which can be broadly classified as **descriptive** study/survey, **observational** study or randomized experiment.



- ✓ Randomized
- ✓ Controlled

Types of statistical studies

Controlled experiment

In a **controlled experiment**, participants are randomly assigned to participate in a **specific treatment** or a **control group** without treatment.

- Try to setup something valid to compare with (**a solid reference**) called a **Control group**.
 - Where all **lurking/confounding** variables or factors that might affect the results are (attempted to be) **controlled**.
- **Lurking variables** are unforeseen third variables that affect observational studies.
- **Confounding variables** are those that affect the response variable and are related to the explanatory/independent variable.

Can meditation improve test scores?



MIND

How Meditation Might Boost Your Test Scores

By JAN HOFFMAN APRIL 3, 2013 12:05 PM 54 Comments

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Mindfulness meditation, the ancient and [flourishing practice](#) that increases awareness of random thoughts and redirects attention to the present moment, has been used to manage stress, depression and even chronic pain. But can it improve test scores?

Researchers in the department of psychological and brain sciences at the University of California, Santa Barbara, who have been studying the relationship between mindfulness and mind-wandering, or the tendency to let our minds drift away on “task-unrelated thoughts,” as it is referred to in academic literature, sought to find out.

“We had already found that mind-wandering underlies performance on a variety of tests, including working memory capacity and intelligence,” said Michael D. Mrazek, a graduate student working with [Jonathan W. Schooler](#), a professor of

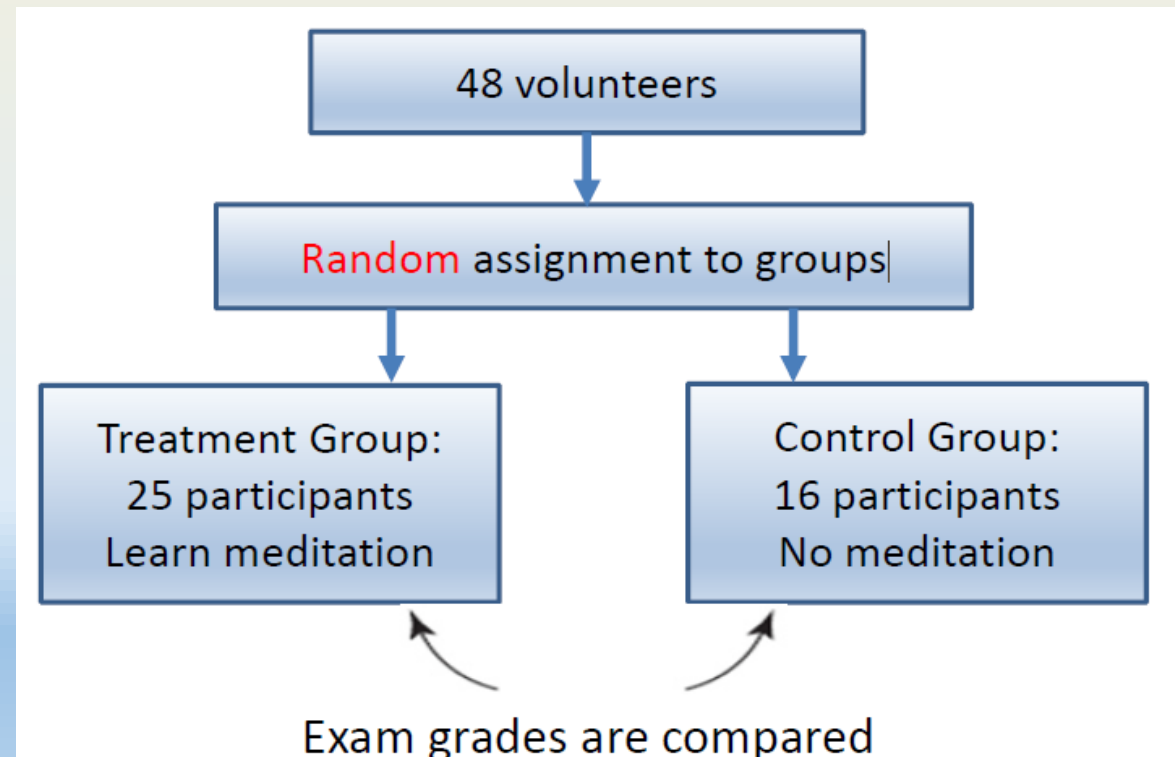
Types of statistical studies

Study example 1 (how it was controlled)

In a **controlled experiment**, participants are randomly assigned to participate in a **specific treatment** or a **control group** without treatment:

- e.g., learn meditation or
- No meditation as follows

A controlled experiment conducted by researchers in the Department of psychological and brain sciences at the University of California, Santa Barbara:



Types of statistical studies

Observational study

- In observational study, researchers only **observe or measure the participants** and do not assign any treatments or conditions.
- Looks for relations between variables

Types of statistical studies

Study example 2

Can drinking coffee help you live longer?



National Institutes of Health
Turning Discovery Into Health

Embargoed for Release: Wednesday, May 16, 2012, 5 p.m. EDT

NIH study finds that coffee drinkers have lower risk of death

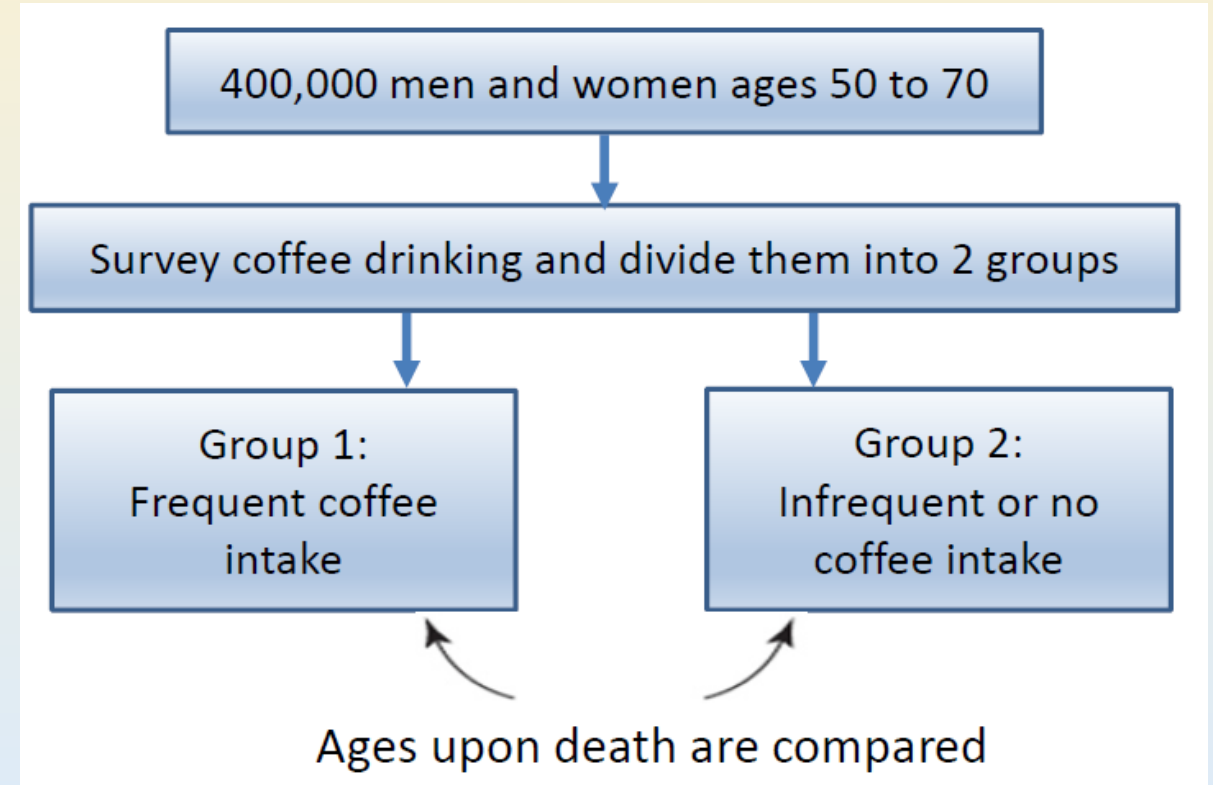
Older adults who drank coffee — caffeinated or decaffeinated — had a lower risk of death overall than others who did not drink coffee, according a study by researchers from the National Cancer Institute (NCI), part of the National Institutes of Health, and AARP.

Coffee drinkers were less likely to die from heart disease, respiratory disease, stroke, injuries and accidents, diabetes, and infections, although the association was not seen for cancer. These results from a large study of older adults were observed after adjustment for the effects of other risk factors on mortality, such as smoking and alcohol consumption.

Types of statistical studies

Study example 2 (how it was observed)

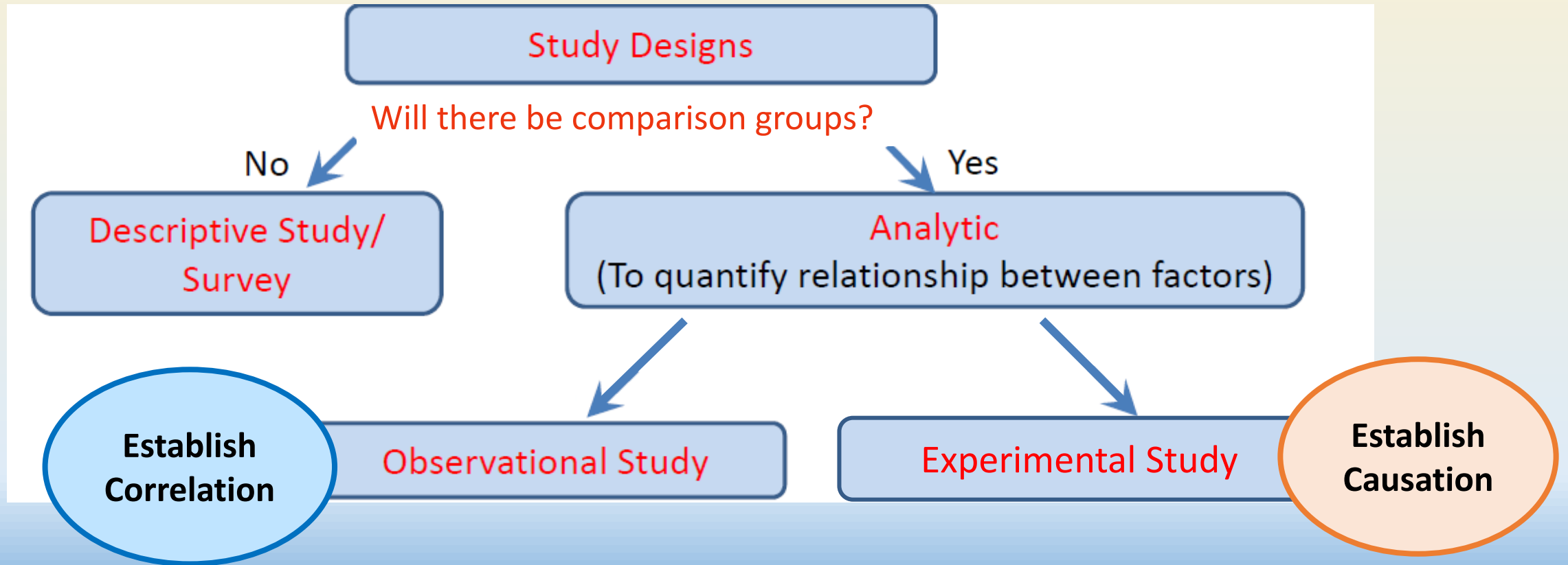
- In observational studies, researchers only observe or measure the participants and do not assign any treatments or conditions.
- e.g. observing coffee intake as follows:



Types of statistical studies

What type of study to use? (Conclusion)

Next, decide on the type of statistical studies to be conducted, which can be broadly classified as descriptive study/survey, observational study or randomized experiment.



CORRELATION VS CAUSATION

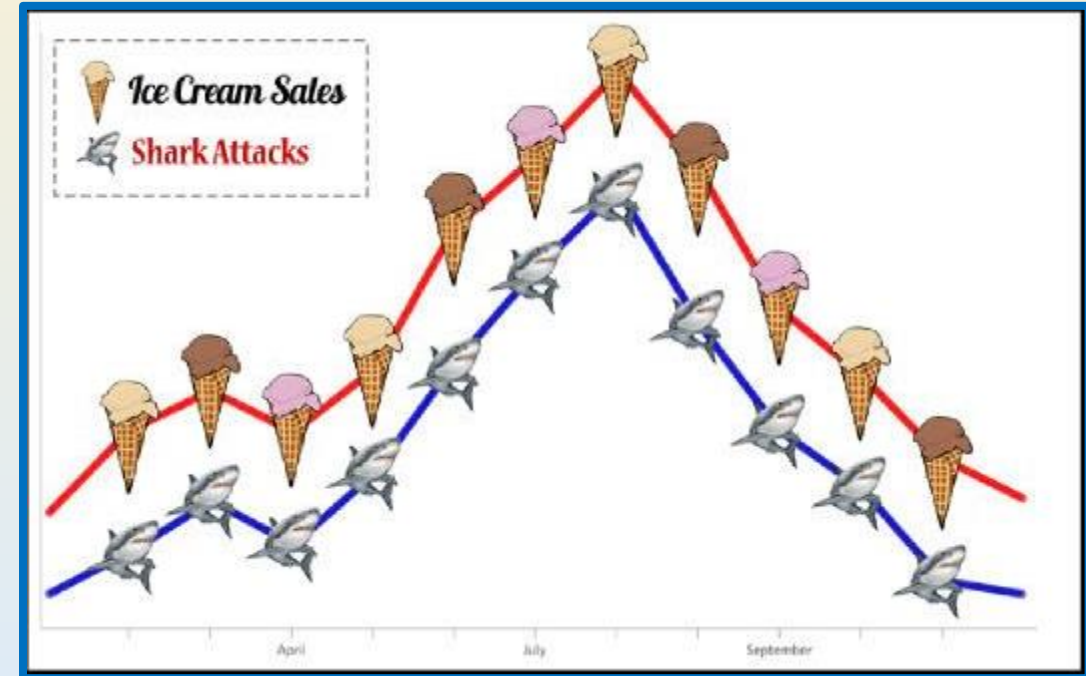
EXAMPLE

Correlation vs Causation

Ice cream shark example

“As ice cream sales increase, there is a correlation in the increased rate of shark attacks.”

Should we restrict the number of ice-cream being sold to reduce the likelihood of a shark attack?



Source <https://bit.ly/2Foktt7>

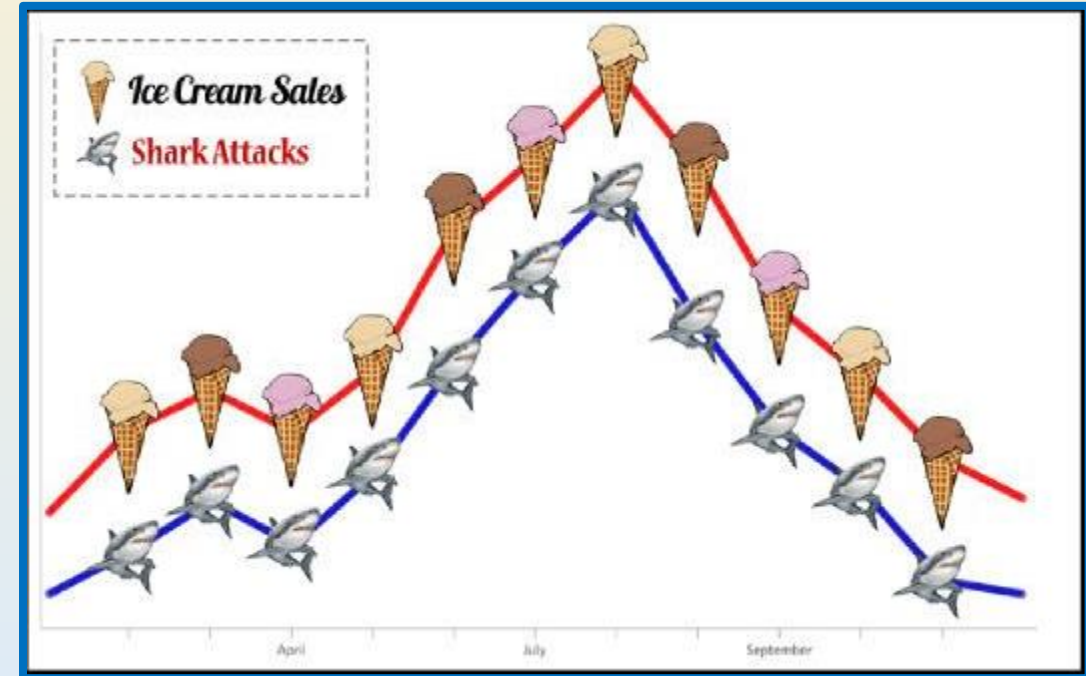
Correlation vs Causation

Ice cream shark example

“As ice cream sales increase, there is a correlation in the increased rate of shark attacks.”

Should we restrict the number of ice-cream being sold to reduce the likelihood of a shark attack? **(NO)**

The rise in **shark attacks** is not caused by ice cream sales, but by the unstated factor (a **lurking variable**) of **warm weather** sending more people to the beach



Source <https://bit.ly/2Foktt7>

Correlation vs Causation

Ice cream shark example

Correlation **doesn't imply** Causation

“Correlation only means that the two events co-exist more often than ordinary chance. They are changing together somehow.”



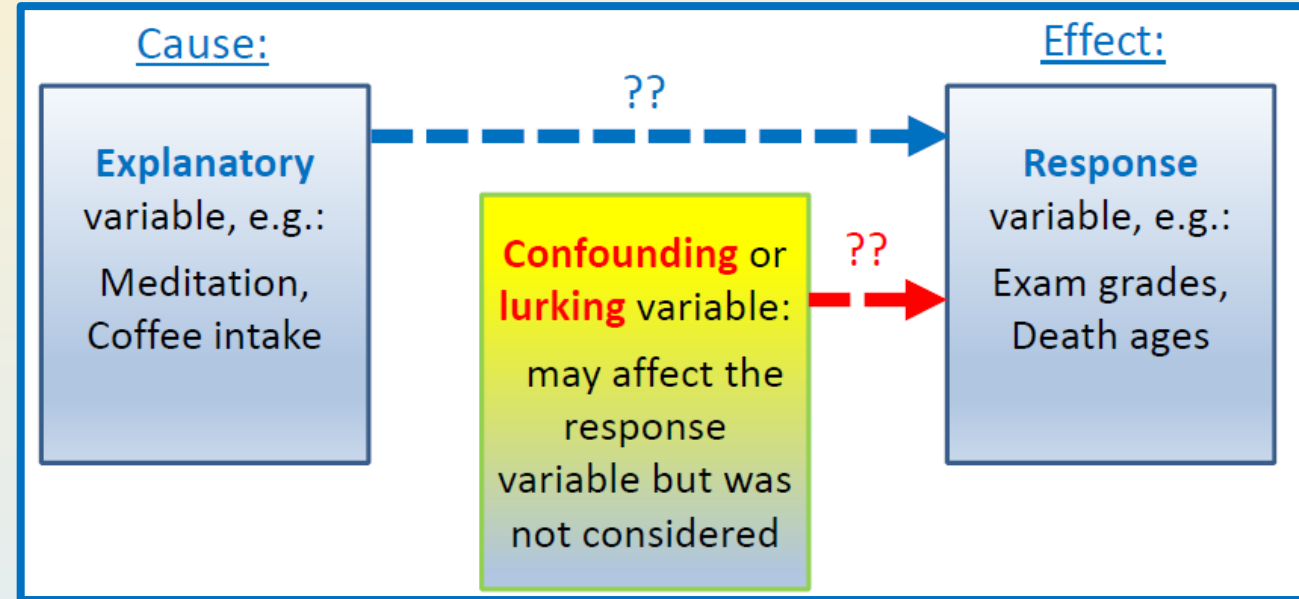
*Loving those
ice-cream
flavored humans*

Source <https://bit.ly/2Foktt7>

Correlation vs Causation

Why Correlation is not Causation

- Note that unlike **controlled experiments**, cause-and-effect relationship usually **CANNOT** be inferred from **observational studies**! (Most common mistake made by many.)
- Many relationships are complex, and it is unlikely that any explanatory variable is the direct and sole cause for the results of response variable:
- **Confounding/Lurking** variables are less likely to be problems for randomized experiment due to random assignment of participants which should approximately even out the influence in the two groups.



Samples & Sampling distributions

SUMMARY of studies

Descriptive research or sample survey

- You randomly sample and collect data through surveys (or similar means).

Observational Studies

- A researcher uses observed information to learn about a group of individuals.
- The individuals being observed are specifically **not interfered** with aside from the measuring of their responses
- Such study doesn't establish causality because there could be confounding or lurking variables but can help establish **correlation**.

Randomized Controlled Experiment

- A researcher deliberately imposes different treatments on different groups and then compares the groups to determine if there is any difference in a specific variable of interest (called a response).
- Researchers can **establish causality** through this.